

The logo features the word "RAOH" in a large, bold, white sans-serif font. The background is a dark green with faint, large-scale geometric patterns, including a large "R" shape on the left and a series of curved lines at the bottom right.

RAOH

— R e s e a r c h e r s —

RESEARCH. DISCOVER. IMPACT.

Advancing knowledge.
Creating real-world impact.

Research Fundamentals

Key Pillars of a Strong Medical Study

In medical research, a good idea alone is not enough.

A strong study is built on clear scientific foundations that determine whether the results are meaningful, reliable, and publishable.

These foundations include:

- Research Question
- Sample
- Validity
- Bias & Confounders

Understanding these elements is essential before designing or interpreting any study.

1. Research Question

The Starting Point of Every Study

Every research project begins with a clear and focused question.

Examples:

- Do medical students experience high levels of stress?
- Is sleep duration associated with academic performance?

A weak or unclear question leads to a weak study, no matter how good the analysis is.

What Makes a Good Research Question?

A strong research question depends on three key criteria:

1. Novelty (Is it new?)

A research question does not need to be completely original, but it must add scientific value.

Ways to evaluate novelty:

- Search in **Google Scholar**
- Check if the topic has already been studied in the same population and setting

Types of novelty:

- **Exact repetition (low value):**
Same population, same country, same variables → usually redundant
- **Replication study (acceptable):**
Same topic but different population or country
- **Local gap study (high value):**
Topic not studied in your region (e.g., Jordan) → very strong for student research

👉 In research, value does not come from “inventing something new,” but from adding meaningful evidence.

2. Measurability

A good research question must be measurable and convertible into data.

Weak example:

- “Stress among students”

Strong example:

- “Prevalence of stress among medical students using a validated scale”

The second one is measurable, structured, and scientifically usable.

3. Feasibility

A research idea must be practically achievable.

Key questions:

- Can you reach the target population?
- Is the sample accessible?
- Is the time sufficient?
- Are resources available?

Even a perfect idea is useless if it cannot be executed.

2. Sample

The Population Representation

The sample is a smaller group selected from the larger population.

Example:

To study university students, you cannot survey everyone, so you select a sample that represents them.

Core Principle

A good sample = representative of the real population with minimal bias.

Sampling Methods

1. Random Sampling (Best scientifically)

Every individual has an equal chance of selection.

2. Convenience Sampling (Common in student research)

Selecting easily accessible participants (e.g., classmates).

- Easier
 - Faster
 - Less accurate statistically
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Sample Size

Sample size is not random.

It depends on:

- Population size
- Confidence level (usually 95%)
- Margin of error (usually 5%)

Typical student research range:

200 – 400 participants (approximate, depending on study design)

👉 Key idea:

It is not about having a large sample, but a **sufficient and representative one**.

3. Validity

Are We Measuring What We Think We Are Measuring?

Validity refers to the accuracy of a study.

In simple terms:

Does the study actually measure what it claims to measure?

Types of Validity

1. Content Validity

Does the tool cover all relevant aspects of the concept?

Example: Measuring stress should include:

- Emotional symptoms
 - Physical symptoms
 - Behavioral changes
 - Cognitive aspects
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2. Construct Validity

Does the tool truly measure the theoretical concept?

Example:

Does a depression scale measure actual depression, or just temporary sadness?

3. Face Validity

Does the tool look logically correct at first glance?

Even without deep analysis, it should make sense scientifically.

Key Principle

If the measurement tool is wrong, the entire study becomes unreliable.

4. Research Instruments

Tools for Data Collection

Common instruments include:

- Questionnaires
- Standardized scales
- Interviews

Examples of validated tools:

- PHQ-9 → Depression
 - GAD-7 → Anxiety
 - PSQI → Sleep quality
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Golden Rule for Students

Do NOT invent your own questionnaire as a beginner.

Always use validated tools because they are:

- Scientifically tested
 - Reliable (consistent results)
 - Valid (measure what they are supposed to measure)
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Reliability

Reliability means:

If the measurement is repeated, it should produce similar results.

A tool can be reliable but not valid—but a valid tool must be reliable.

5. Bias

Systematic Error in Research

Bias refers to any error in study design or execution that distorts results.

Main Types of Bias

1. Selection Bias

Occurs when the sample is not representative.

Example:

Only selecting high-achieving students → unrealistic results.

2. Information Bias

Occurs when collected data is inaccurate.

Example:

Participants lying or misreporting behaviors.

3. Recall Bias

Occurs when participants cannot accurately remember past events.

Example:

“How many energy drinks did you consume last month?”

4. Confounding Bias

The most important type.

Occurs when a third variable influences both the exposure and outcome.

Example:

Coffee → Insomnia

But the real factor may be stress.

How to Reduce Bias

- Use proper sampling methods
 - Use validated instruments
 - Design clear questions
 - Apply correct statistical methods
 - Control confounders during analysis
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6. Confounders

Hidden Influencing Factors

Confounders are external variables that affect the relationship between studied variables.

Example

Study question:

Does coffee cause insomnia?

But:

- Stress increases coffee consumption
- Stress also reduces sleep quality

So stress is a confounder.

Common Confounders in Student Research

Always consider:

- Age
 - Gender
 - Year of study
 - GPA
 - Sleep duration
 - Smoking status
 - Coffee intake
 - Physical activity
 - Socioeconomic status
 - Working status
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Why Are Confounders Important?

Because they help interpret results correctly and avoid false conclusions.

Ignoring confounders can completely distort study findings.

7. What Makes a Strong Research Study?

A strong study must include:

1. Clear Research Question

Specific, focused, and measurable.

2. Representative Sample

Adequate and unbiased.

3. Validated Instruments

Reliable and scientifically tested tools.

4. Minimal Bias

Controlled during design and analysis.

5. Controlled Confounders

Identified and adjusted for statistically.

Final Principle

Good research is not defined by complexity, but by clarity, validity, and methodological discipline.